

Java IntStream – One Page Cheat Sheet

Stream Creation

Method	Returns	Time	Boxing	Use
<code>Arrays.stream(int[])</code>	<code>IntStream</code>	$O(1)$	No	From array
<code>IntStream.of()</code>	<code>IntStream</code>	$O(n)$	No	From values
<code>range / rangeClosed</code>	<code>IntStream</code>	$O(1)$	No	Ranges
<code>generate / iterate</code>	<code>IntStream</code>	$O(1)$	No	Infinite streams

Intermediate Operations

Method	Returns	Time	Boxing	Notes
<code>filter</code>	<code>IntStream</code>	$O(n)$	No	Predicate
<code>map / flatMap</code>	<code>IntStream</code>	$O(n)$	No	Transform
<code>distinct</code>	<code>IntStream</code>	$O(n)$	No	Hash-based
<code>sorted</code>	<code>IntStream</code>	$O(n \log n)$	No	Primitive sort
<code>limit / skip</code>	<code>IntStream</code>	$O(k)/O(n)$	No	Short-circuit

Terminal Operations

Method	Returns	Time	Boxing	Use
<code>sum</code>	<code>int</code>	$O(n)$	No	Total
<code>min / max</code>	<code>OptionalInt</code>	$O(n)$	No	Min / Max
<code>average</code>	<code>OptionalDouble</code>	$O(n)$	No	Mean
<code>count</code>	<code>long</code>	$O(n)$	No	Count
<code>summaryStatistics</code>	<code>Stats</code>	$O(n)$	No	All stats
<code>reduce</code>	<code>OptionalInt/int</code>	$O(n)$	No	Custom reduce

Conversion

Method	Returns	Time	Boxing	Warning
<code>boxed</code>	<code>Stream<Integer></code>	$O(n)$	YES	Heap allocation
<code>mapToLong</code>	<code>LongStream</code>	$O(n)$	No	Safe
<code>mapToDouble</code>	<code>DoubleStream</code>	$O(n)$	No	Safe